

Subject: Application for Java Software Engineer (AI) – Adrian Vrabie

Dear Sygnum Bank Hiring Team,

I am writing to express my strong interest in the Java Software Engineer position at Sygnum Bank. With a Master's in Mathematics and over a decade of experience in engineering high-performance distributed systems in regulated environments — most recently at UBS — I am eager to contribute to Sygnum's mission of building production-grade AI systems that are governed, compliant, and reliable.

I have been actively exploring the intersection of Java and Artificial Intelligence, primarily through hackathons and intensive prototyping. During a recent two-day challenge, I developed a working POC for a document classification algorithm using **Spring AI** and **Embabel**. The system was designed to automatically detect document types and identify sensitive Client Information (CI). This project involved designing end-to-end **RAG pipelines**, where I experimented with various embedding models and utilized **PGVector** for similarity search.

I am particularly excited about your focus on moving AI from POC to production. While my AI implementation experience has been focused on rapid prototyping at hackathons, my decade of experience at firms like UBS and Zurich Insurance has taught me exactly what it takes to transition a working model into a governed, compliant, and reliable product. I understand that in a regulated industry, success demands rigorous access controls, audit trails, and operational discipline.

Technically, I have deep experience in the Spring ecosystem, with a particular passion for reactive programming using **Spring Boot WebFlux** and **Project Reactor**. I believe non-blocking architectures are superior for modern AI applications, especially when handling long-lived LLM chat streams or complex “online” algorithms where latency and resource efficiency are critical. My background also includes strong experience with microservices, containerization (Docker, Kubernetes), and cloud platforms like Azure.

Furthermore, I have been exploring **Agent design patterns** (ReAct, multi-agent orchestration) and the **Model Context Protocol (MCP)**, allowing me to choose the right architecture for the problem at hand rather than relying on a single framework.

I am confident that my combination of deep Java expertise, hands-on AI prototyping experience, and familiarity with the requirements of regulated banking environments makes me an ideal fit for Sygnum Bank. I look forward to the possibility of discussing how I can help your team ship and operate world-class AI systems.

Thank you for your time and consideration.

Best regards,

Adrian Vrabie +41 762575204 mr.vrabie@gmail.com <https://blog.s4v3.net>